Supplementary Material for “A Homogeneous Compensation Mechanism for Indirect Evolutionary Rescue in a Roy-Style Predator–Prey Model”

Draft author list to be finalized

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S1. Numerical classification protocols

The main manuscript reports consolidated outputs from the numbered analysis scripts rather than new simulations. Classification used the same vocabulary across ODE and PDE outputs:

- **Persistent**: predator density remains positive and late-window change/residual diagnostics indicate a steady or effectively settled state.
- **Extinct**: predator density approaches the extinction threshold and late-window diagnostics do not indicate recovery.
- **Transient**: the trajectory does not meet persistent or extinct steady criteria within the tested horizon, often because late-window change or residual diagnostics remain non-negligible.

The tail-window logic was used to avoid treating long transients as equilibria. For PDE tests, classification of spatial means was paired with spatial coefficient-of-variation diagnostics for n, w, and q. A basin label was compared with a matched homogeneous control when heterogeneous perturbation effects were evaluated. The classification logic is diagnostic, not a proof of global basin structure.

S2. Mathematical appendix: ODE Jacobian and Routh–Hurwitz margins

Write

$$F_n = nR, \quad R = r(q)z - \xi - a(q)w,$$

$$F_w = wP, \quad P = b(q)nz - (m+s) - \mu w,$$

$$F_q = \nu q(1-q)G, \quad G = \Delta r z - \Delta a w,$$

with $z = \kappa^{-1} - n - w$, $\Delta r = r_v - r_u$, $\Delta a = a_v - a_u$, and $\Delta b = b_v - b_u$. Since $z_n = z_w = -1$,

$$R_n = -r(q), \qquad R_w = -r(q) - a(q), \qquad R_q = G,$$

$$P_n = b(q)(z-n), \qquad P_w = -b(q)n - \mu, \qquad P_q = \Delta b\, nz,$$

$$G_n = -\Delta r, \qquad G_w = -\Delta r - \Delta a, \qquad G_q = 0.$$

The full reaction Jacobian in variables (n, w, q) is therefore

$$J = \begin{pmatrix} R - nr(q) & n[-r(q) - a(q)] & nG \\ w\, b(q)(z - n) & P + w[-b(q)n - \mu] & w\, \Delta b\, nz \\ -\nu q(1-q)\Delta r & -\nu q(1-q)(\Delta r + \Delta a) & \nu(1-2q)G \end{pmatrix}.$$

At an interior compensation-branch equilibrium, $R = P = G = 0$, giving the simplified branch Jacobian

$$J_* = \begin{pmatrix} -n^* r(q^*) & -n^*[r(q^*) + a(q^*)] & 0 \\ w^* b(q^*)(z^* - n^*) & -w^*[b(q^*)n^* + \mu] & w^* \Delta b\, n^* z^* \\ -\nu q^*(1-q^*)\Delta r & -\nu q^*(1-q^*)(\Delta r + \Delta a) & 0 \end{pmatrix}.$$

For $\det(\lambda I - J) = \lambda^3 + A_1\lambda^2 + A_2\lambda + A_3$, the coefficients were computed as

$$A_1 = -\mathrm{tr}(J), \qquad A_2 = \frac{1}{2}\left[(\mathrm{tr}J)^2 - \mathrm{tr}(J^2)\right], \qquad A_3 = -\det(J).$$

Table 1 reports the Routh–Hurwitz terms and the minimum margin at the target stresses.

Table 1: Routh–Hurwitz margins for the linear compensation branch at target stresses. Values are copied to `results/roy_ode_compensation_routh_hurwitz_margins.csv`.

s	A_1	A_2	A_3	$A_1A_2 - A_3$	min margin
0	3.9468	0.33673	0.003540	1.3255	0.003540
0.069448242	4.3241	0.27528	0.004006	1.1863	0.004006
0.11765625	4.5861	0.21486	0.003566	0.98178	0.003566
0.1584375	4.8077	0.15238	0.002706	0.72989	0.002706
0.16486816	4.8426	0.14158	0.002529	0.68308	0.002529
0.175	4.8976	0.12403	0.002228	0.60525	0.002228

S3. Additional ODE basin maps

The main text focuses on the derived compensation branch and its stability. Additional ODE diagnostics support the basin interpretation and the homogeneous reaction-level mechanism. Figure 2 shows representative ODE time series and selection/growth diagnostics. Figure 3 shows the ODE q_0–w_0 basin map used to relate initial defense and predator density to persistent, extinct, and transient outcomes. Figure 4 shows selection-growth phase diagnostics and numerical equilibrium stability.

S4. PDE non-homogeneous perturbation diagnostics

The main manuscript reports the conclusion from spatial-mode stability and long-horizon basin-resolution analysis. The supporting PDE diagnostics are shown here. Figure 5 summarizes additional spatial-stability views. Figure 6 shows the sampled continuous-λ scan over the mode-equivalent range $i, j = 0, \ldots, 64$, sampled at 600 equally spaced λ values. Figures 7 and 8 show representative non-homogeneous initial and final fields. Figures 9 and 10 show mean dynamics and spatial coefficient-of-variation diagnostics for the targeted non-homogeneous tests. Figures 11–13 show the long-horizon diagnostics for the finite-horizon basin-changing cases.

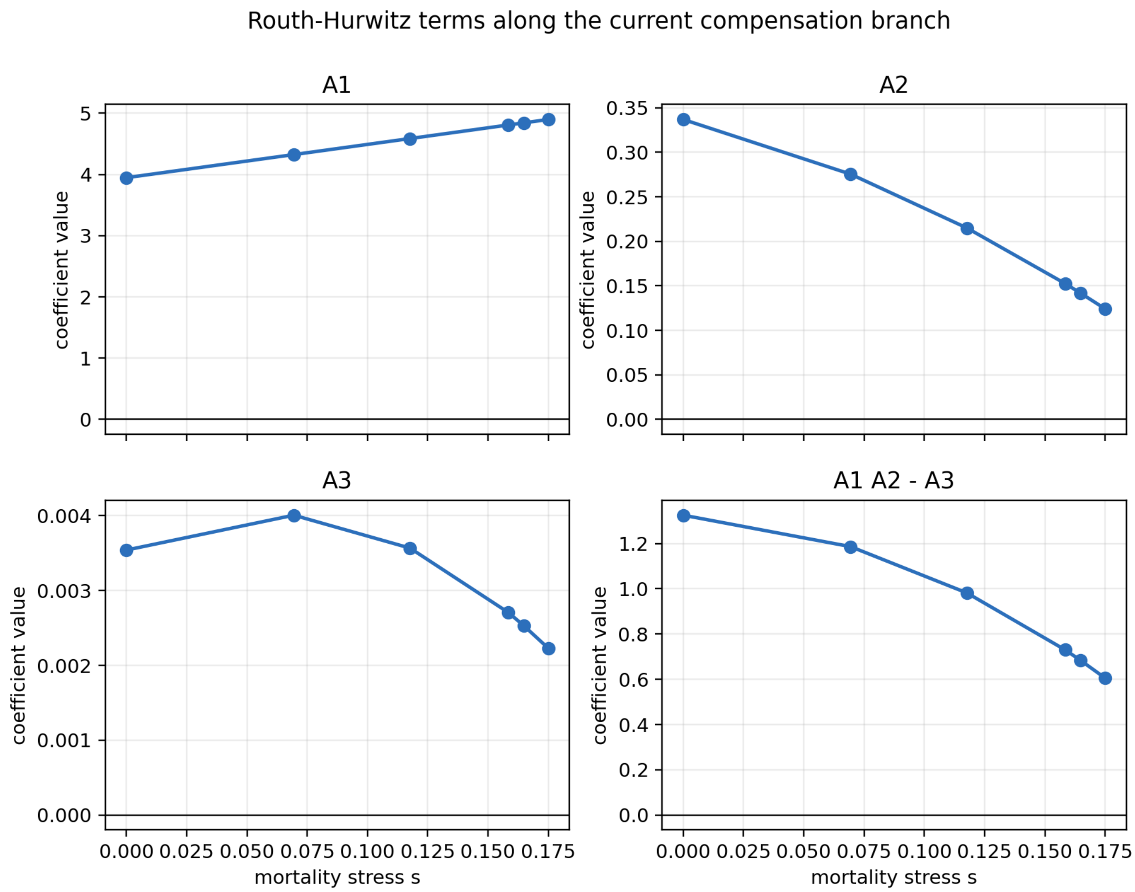

Figure 1: Full Routh–Hurwitz coefficient diagnostics for the linear compensation branch. The main text shows the minimum margin and maximum eigenvalue real part; this supporting figure keeps the individual A_1, A_2, A_3, and $A_1 A_2 - A_3$ terms visible for reproducibility.

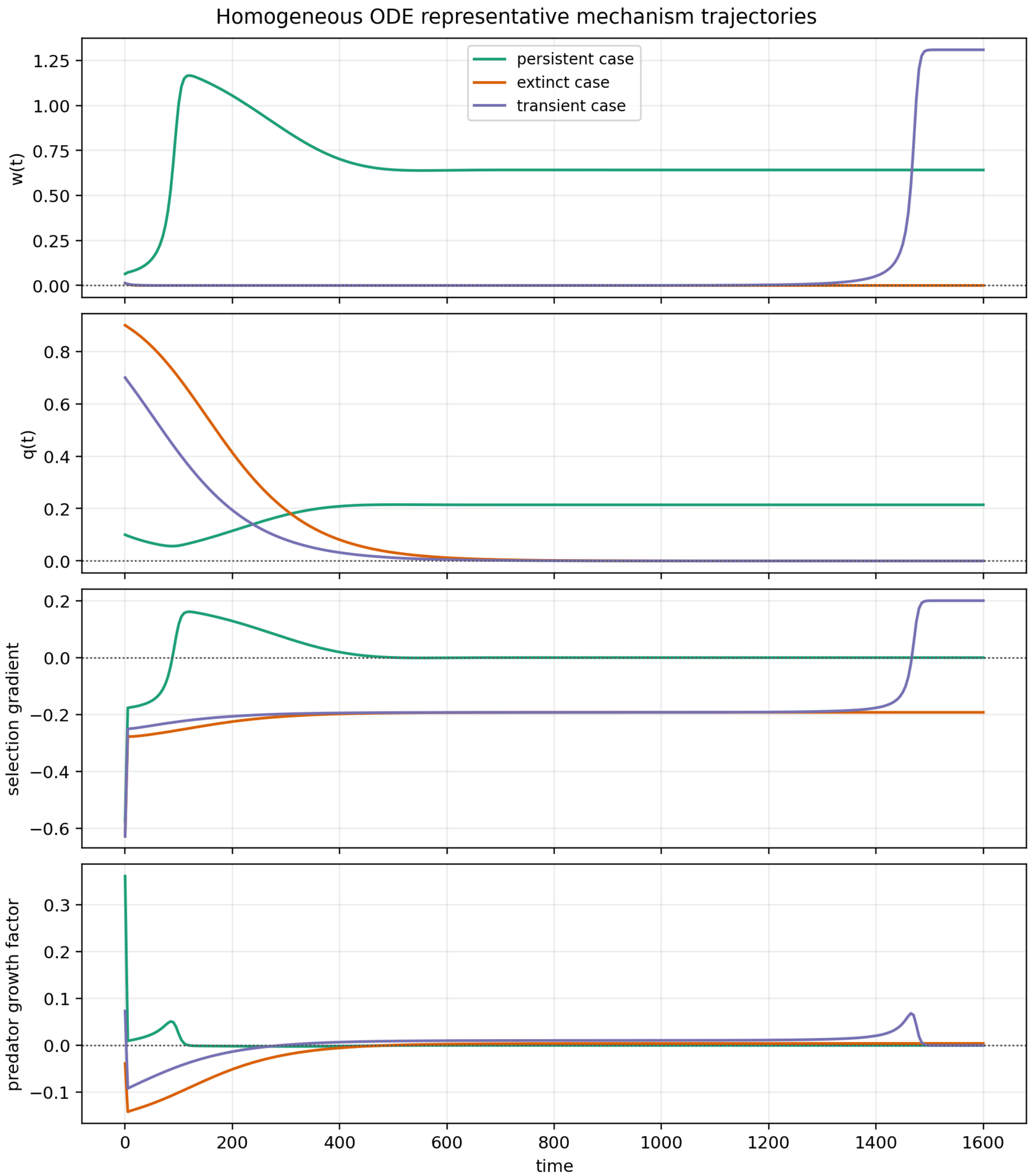

Figure 2: Representative ODE trajectories for predator density, defense frequency, selection-gradient diagnostics, and predator-growth diagnostics. These panels support the homogeneous mechanism interpretation but do not replace the analytic branch derivation.

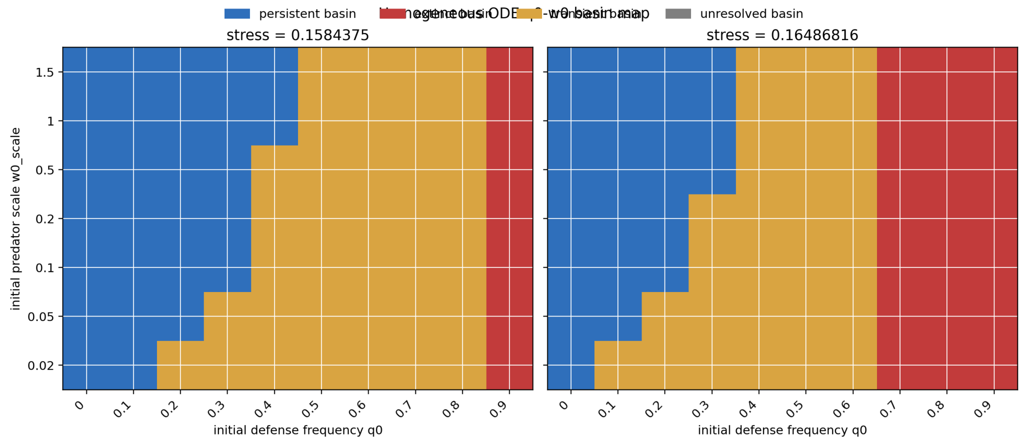

Figure 3: ODE basin map on the targeted q_0–w_0 grid. This figure supports basin dependence in the homogeneous reaction system and is not an exhaustive basin-boundary refinement.

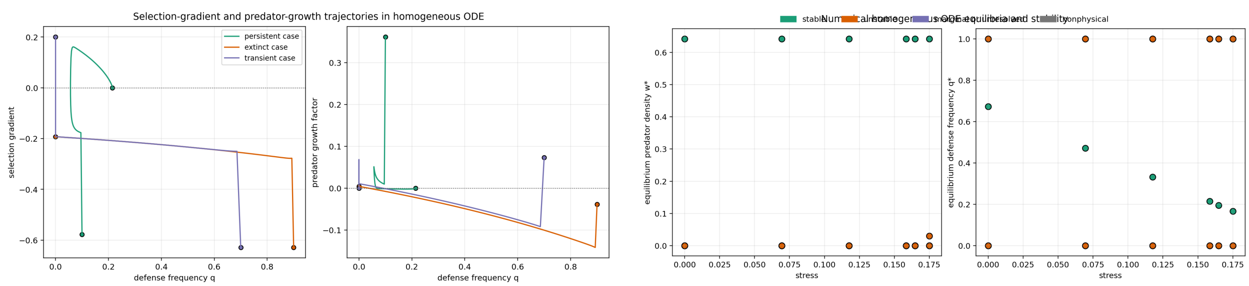

Figure 4: Additional homogeneous mechanism diagnostics. Left: selection-gradient and predator-growth relationships for representative cases. Right: numerical equilibria and stability labels across tested stresses.

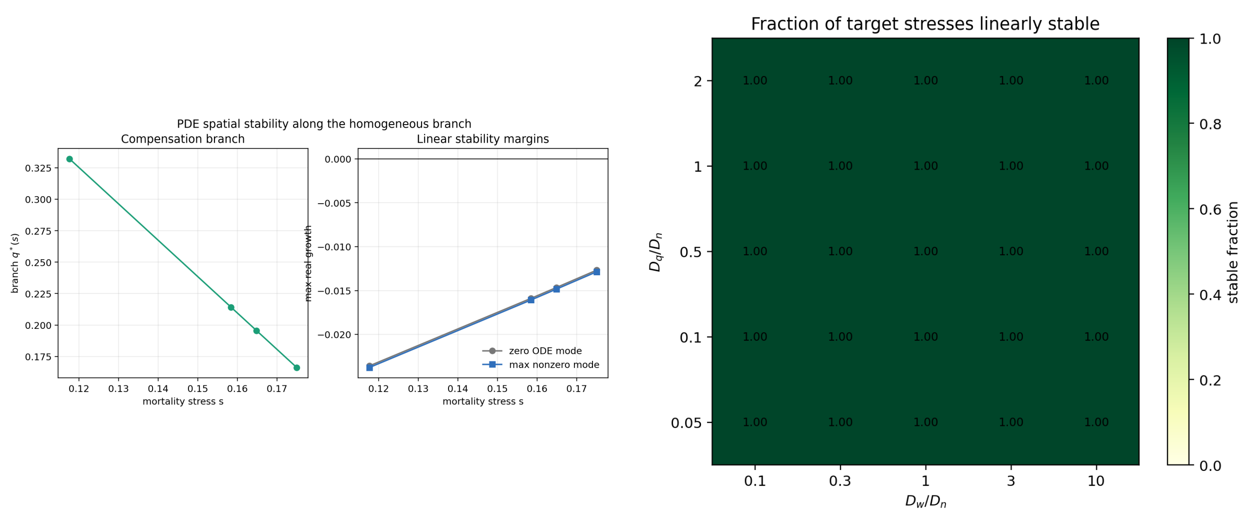

Figure 5: Additional linear PDE spatial-stability diagnostics for the tested branch states and diffusion-ratio grid.

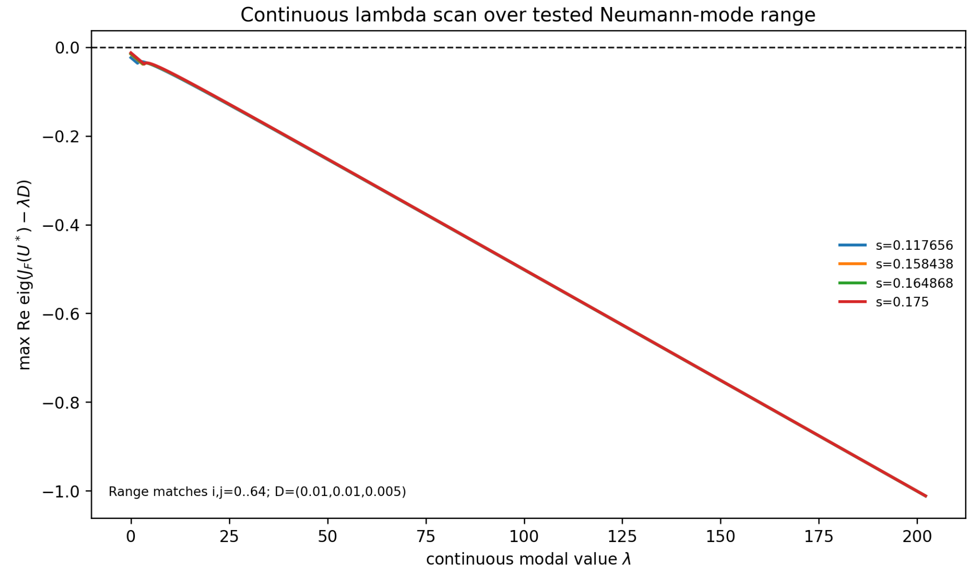

Figure 6: Sampled continuous-λ scan for $J_F(U^*) - \lambda D$ over 600 equally spaced λ values in the mode-equivalent range corresponding to $i, j = 0, \ldots, 64$.

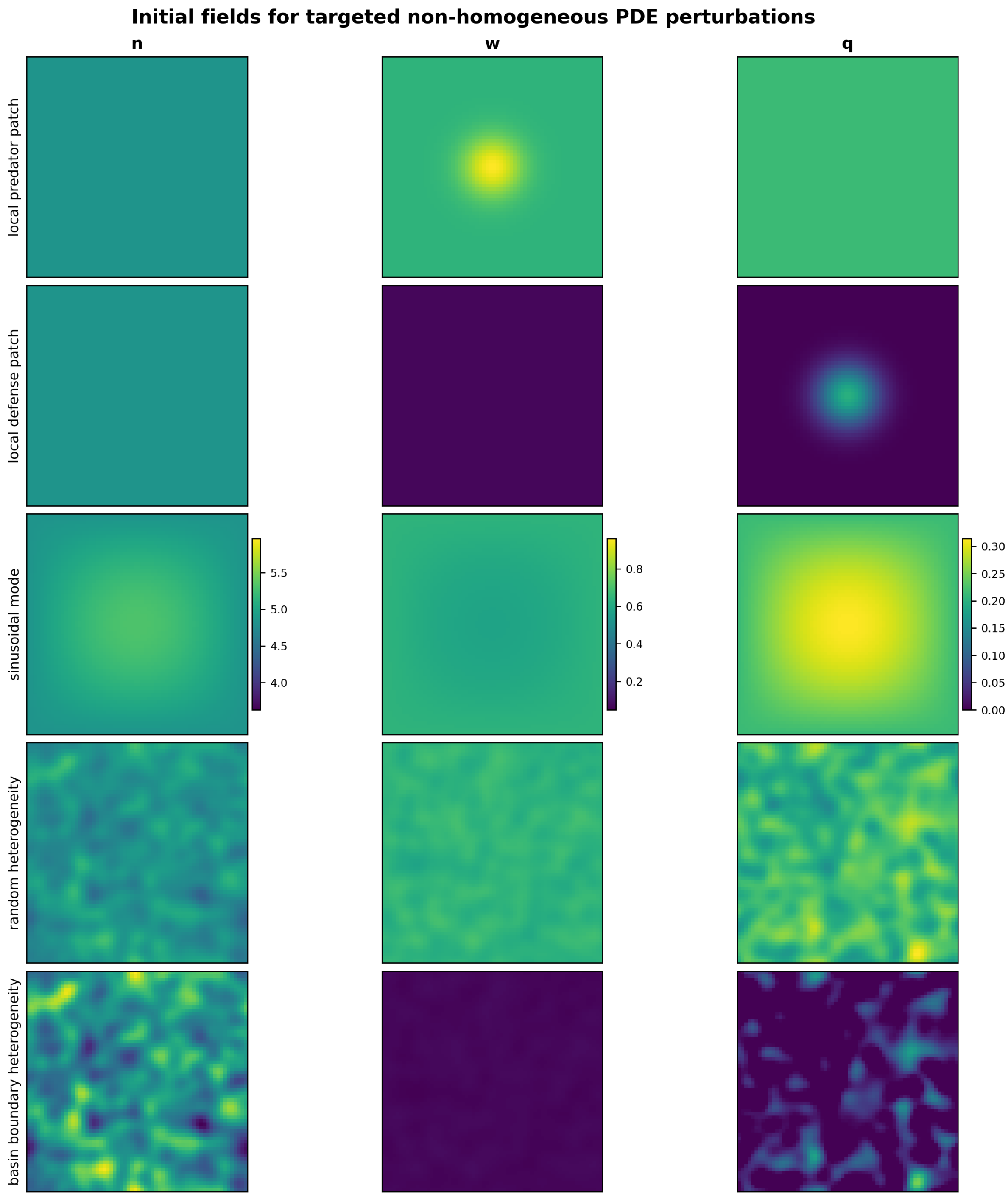

Figure 7: Representative non-homogeneous initial PDE fields for targeted finite-amplitude perturbations. Color scales are variable-specific and shared across rows within this figure; initial and final field figures are scaled independently.

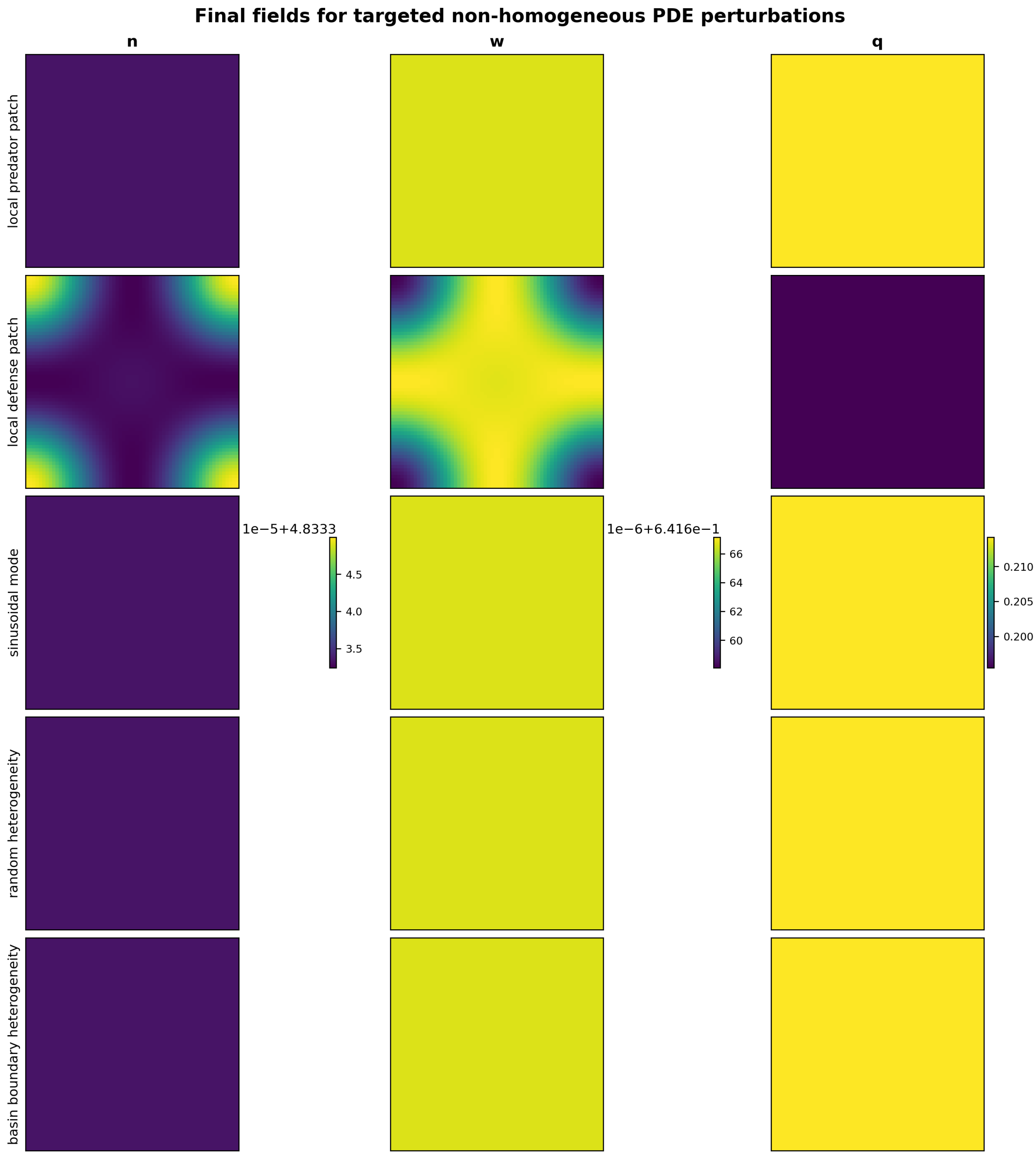

Figure 8: Final PDE fields for representative targeted non-homogeneous perturbation tests. Color scales are variable-specific and shared across rows within this figure; initial and final field figures are scaled independently.

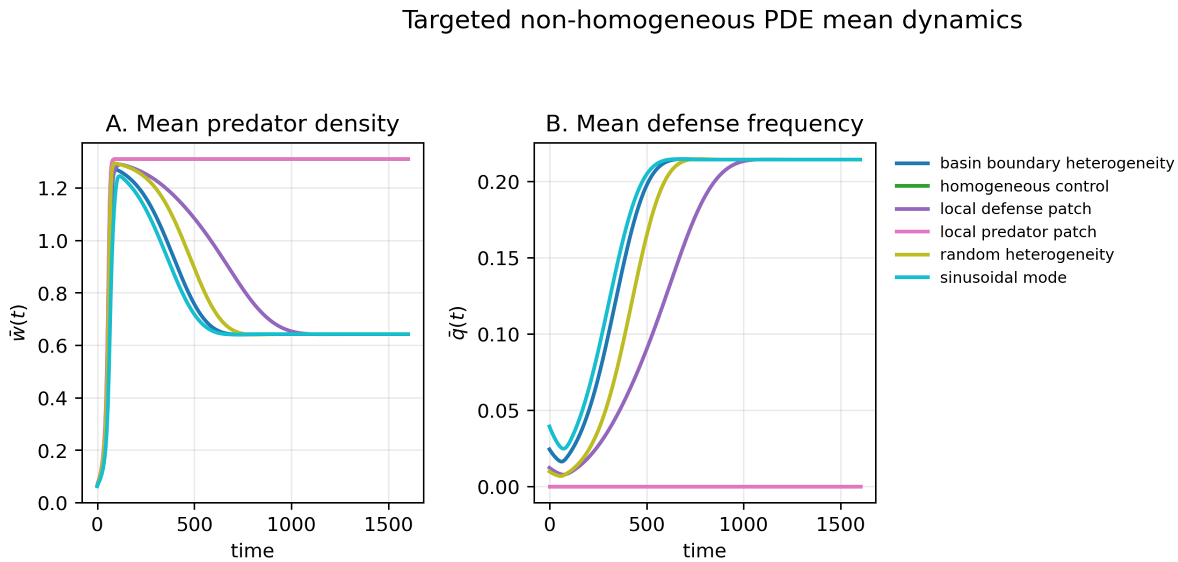

Figure 9: Mean dynamics for targeted finite-amplitude non-homogeneous PDE tests and matched homogeneous controls.

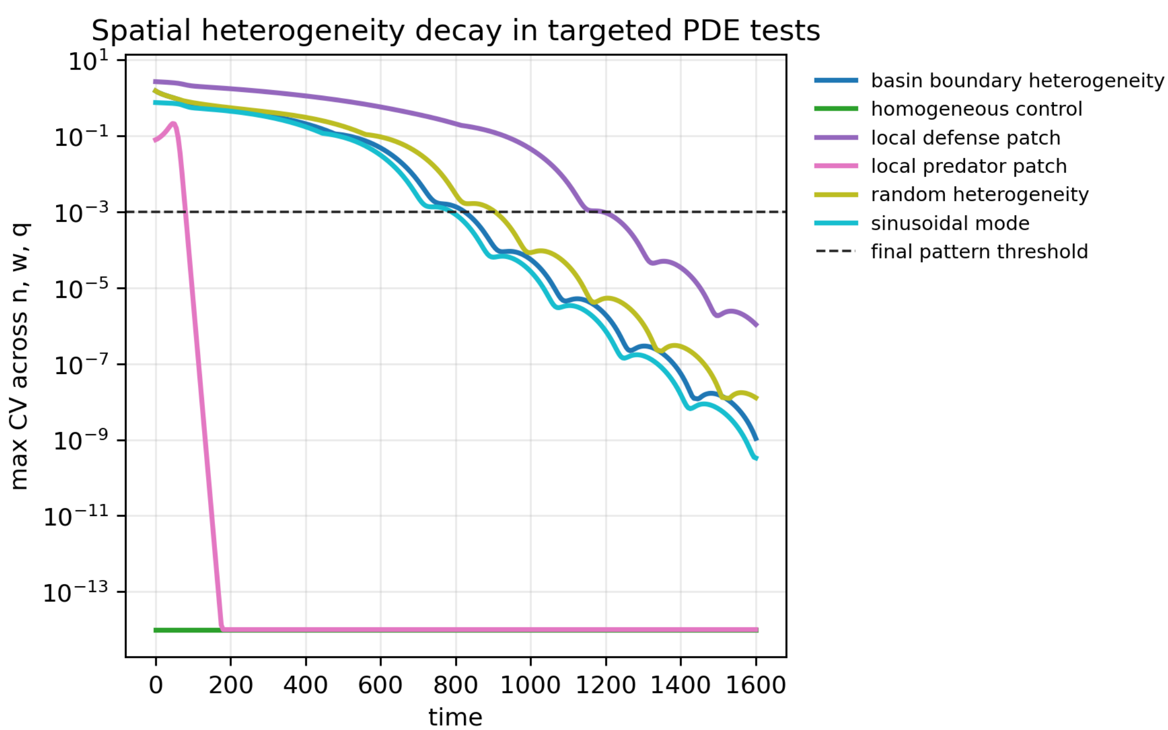

Figure 10: Spatial coefficient-of-variation diagnostics for targeted finite-amplitude non-homogeneous PDE tests.

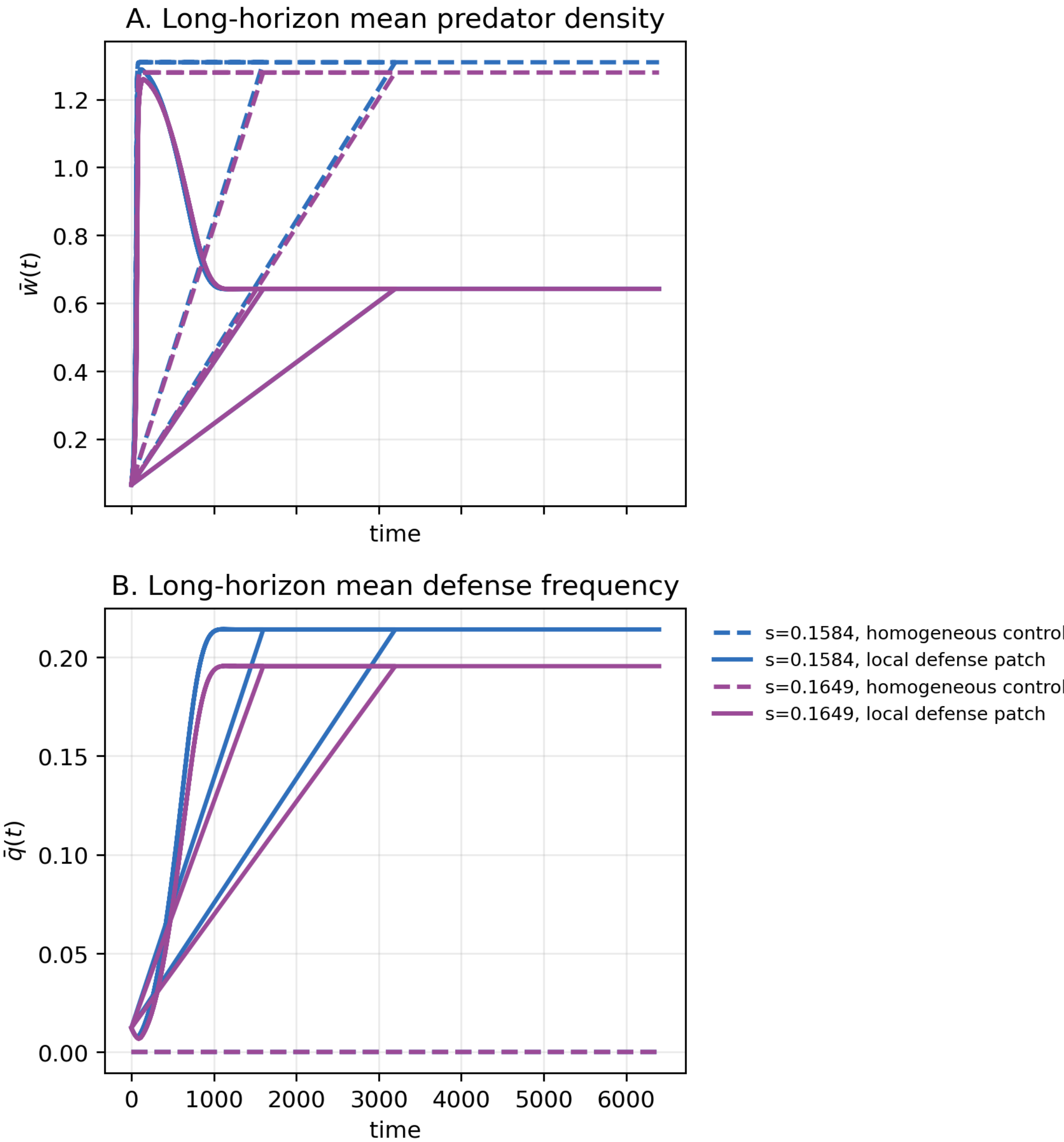

Figure 11: Long-horizon mean time series for the finite-horizon basin-changing non-homogeneous cases. The cases are the targeted cases selected from the earlier finite-amplitude tests, not a broad PDE scan.

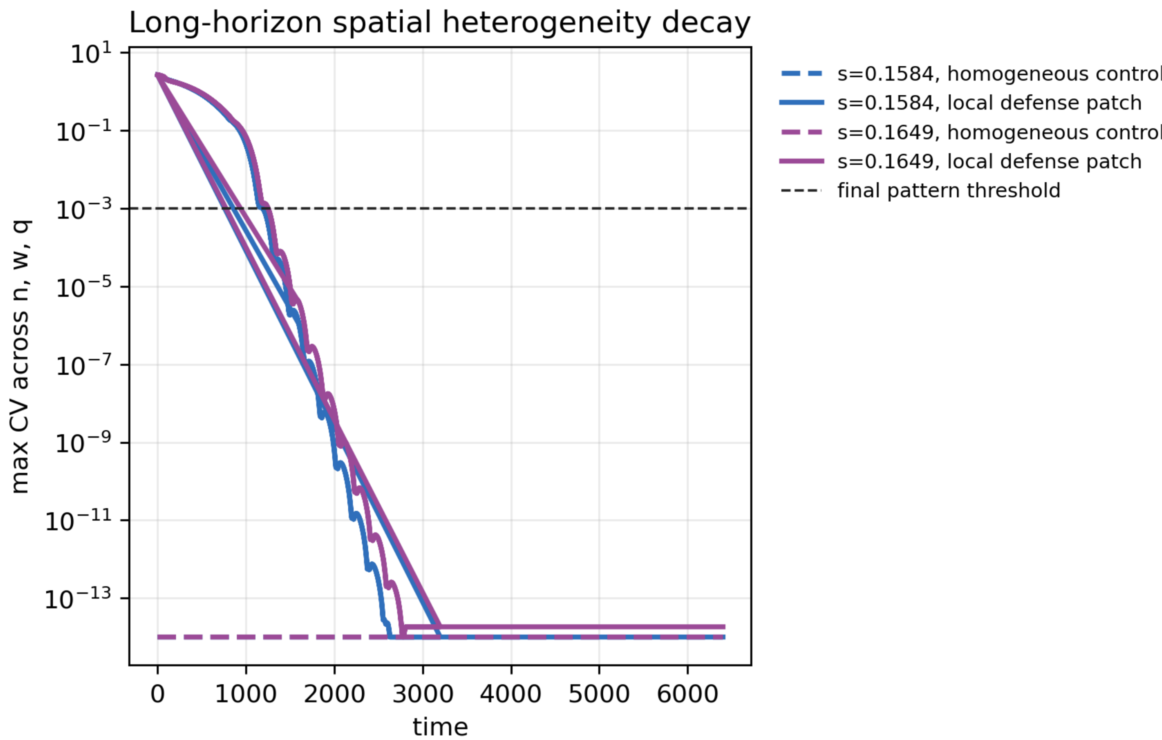

Figure 12: Long-horizon spatial heterogeneity metrics for the followed basin-changing cases. These diagnostics support spatial-pattern decay in the followed cases.

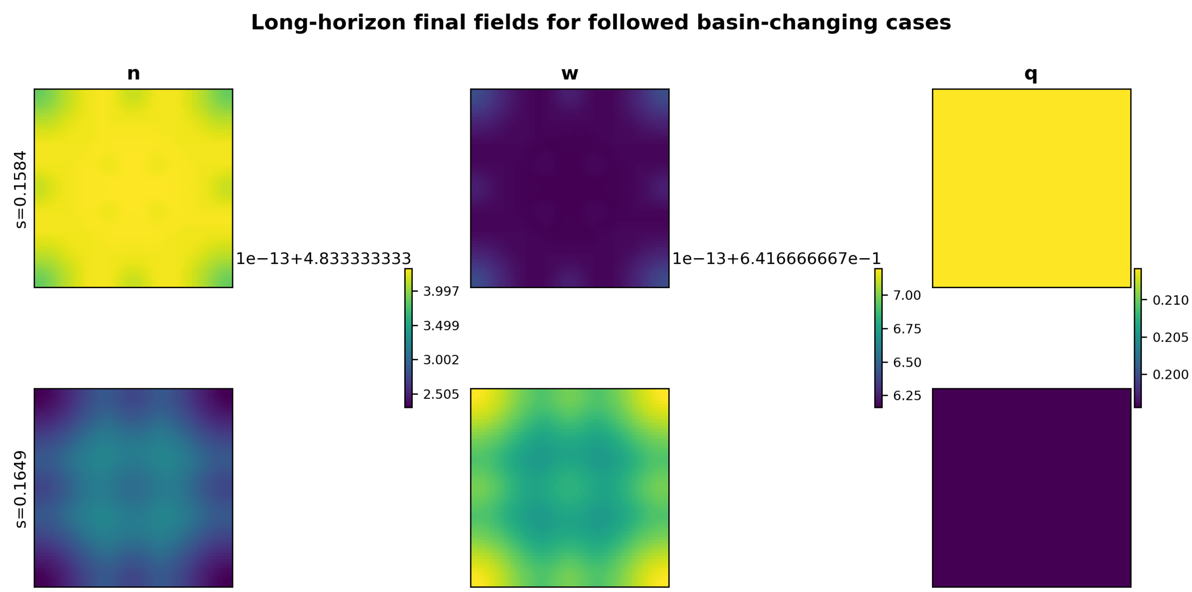

Figure 13: Long-horizon final fields for the basin-changing cases selected for longer-horizon analysis. These diagnostics support resolution without persistent spatial patterning for the followed cases.

S5. Nonlinear trade-off shape-grid details

The nonlinear extension uses endpoint-preserving shape functions $f_\gamma(q) = q^\gamma$ with $\gamma \in \{0.5, 1, 2\}$. The purpose is to test whether the compensation mechanism survives selected concave, convex, and mixed trade-off shapes, not to scan all possible nonlinear functions. Figure 14 shows selected nonlinear branch curves and shape-grid summaries. Figures 15–17 show selected nonlinear ODE basin maps, PDE spatial-stability diagnostics, and non-homogeneous PDE diagnostics.

The no-compensation and unresolved shape classes identify trade-off geometries where the existing diagnostics did not support the compensation mechanism. Table 2 lists these cases from `results/roy_nonlinear_tradeoff_failure_modes.csv`. Four shapes had no feasible branch at the four target stresses. One shape had branches at three target stresses but no target stress was classified as stable in the existing diagnostics. These failure modes show dependence on trade-off geometry; they are not expanded into a broader nonlinear scan here.

Table 2: No-compensation and unresolved nonlinear trade-off shapes from the existing shape-grid diagnostics.

γ_r	γ_a	γ_b	Class	Reason from existing diagnostics
0.5	1.0	0.5	no compensation	No feasible branch at target stresses
0.5	2.0	0.5	no compensation	No feasible branch at target stresses
1.0	2.0	0.5	no compensation	No feasible branch at target stresses
1.0	2.0	1.0	no compensation	No feasible branch at target stresses
2.0	1.0	0.5	unresolved	Branch found at three target stresses, but no stable target stress

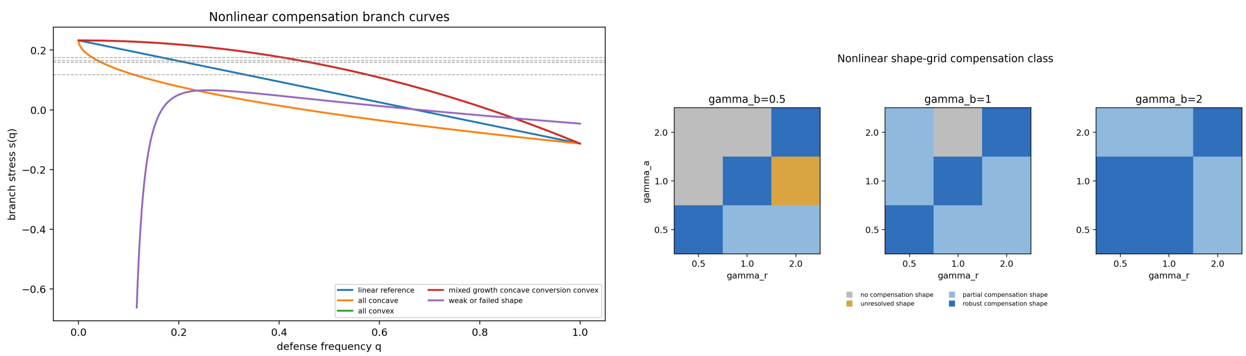

Figure 14: Nonlinear branch curves and shape-grid summary. These figures support selected nonlinear compensation branches within the controlled grid.

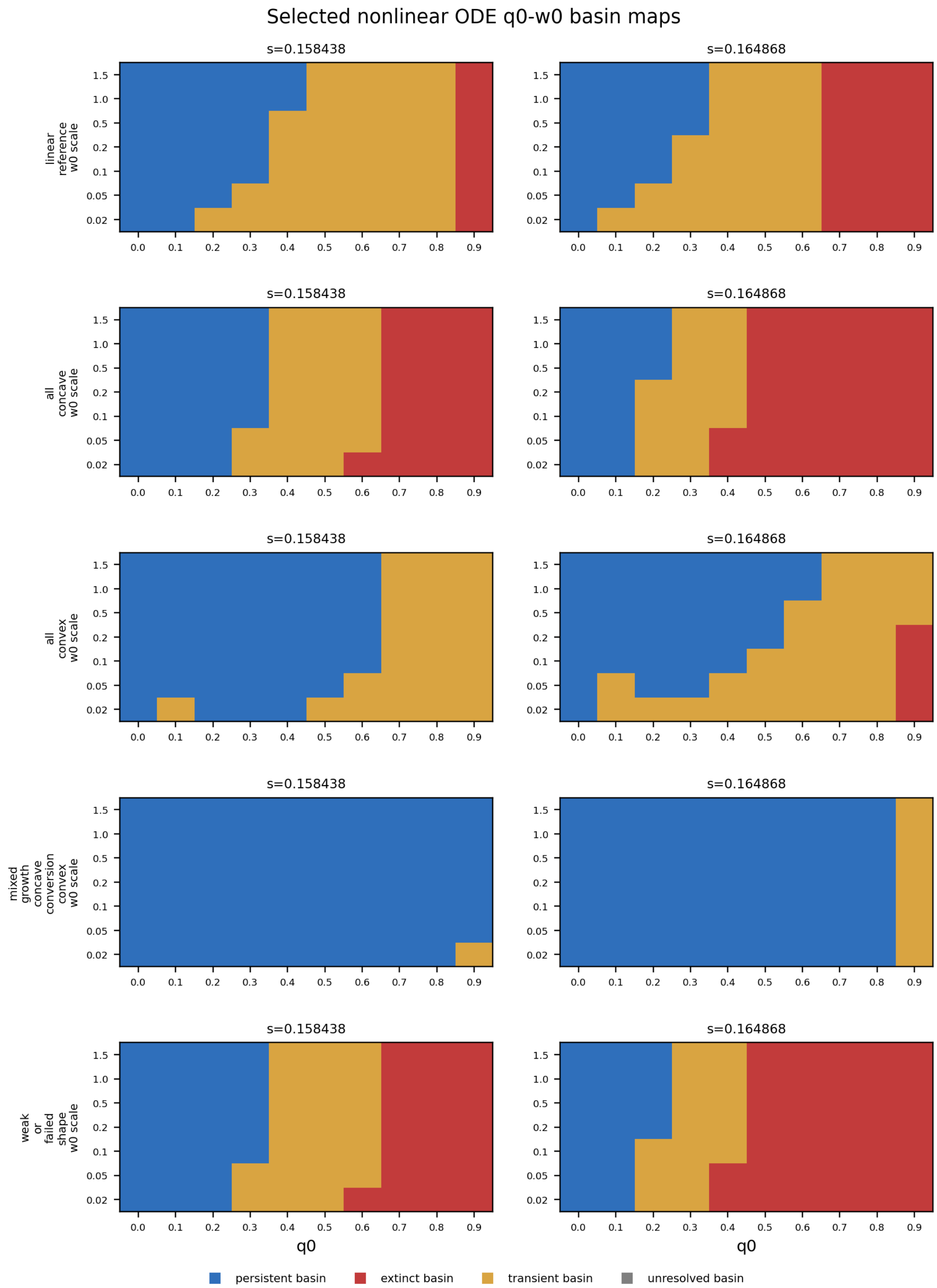

Figure 15: Selected nonlinear ODE basin maps for the controlled shape-grid extension.

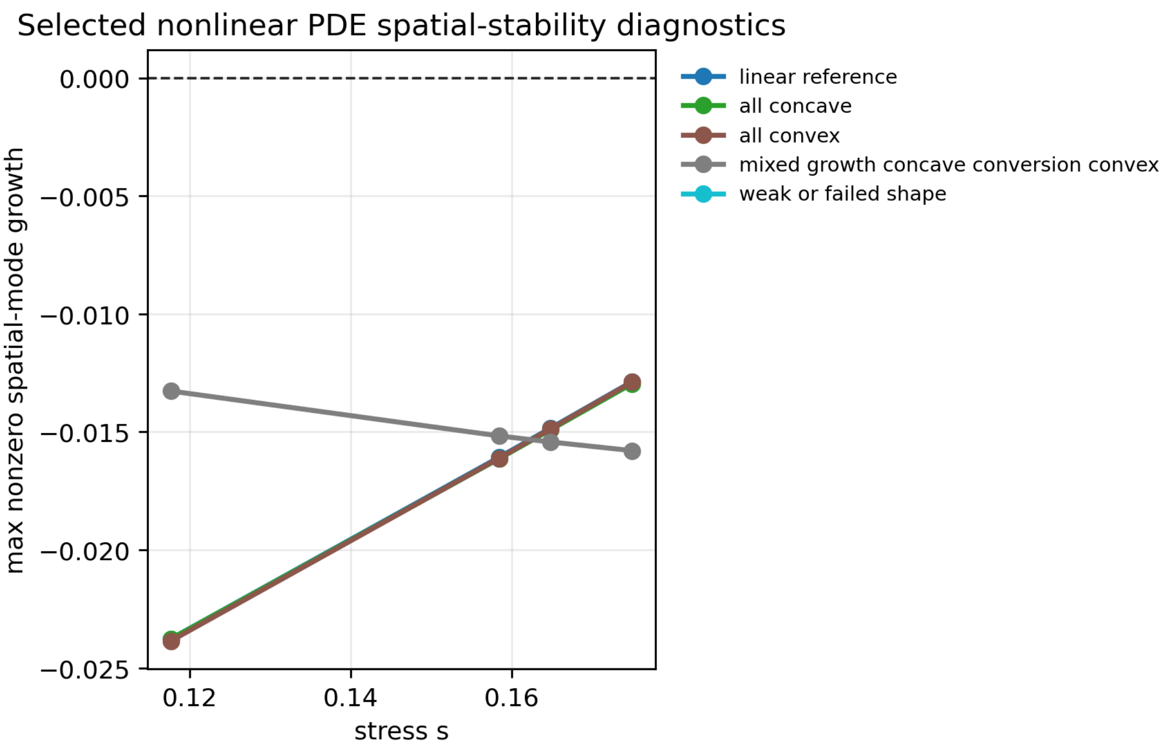

Figure 16: Selected nonlinear PDE spatial-stability diagnostics for selected nonlinear branches.

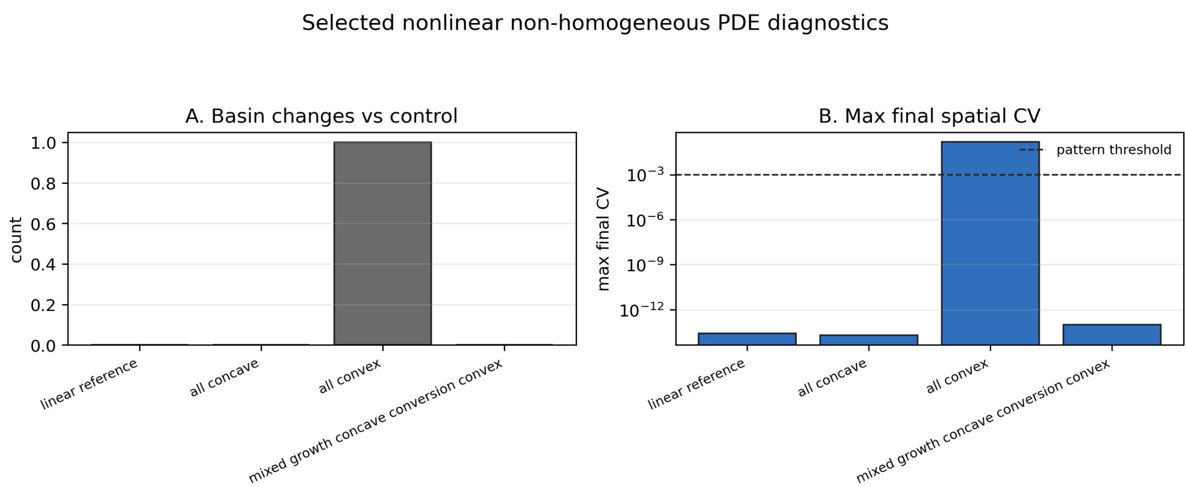

Figure 17: Selected nonlinear non-homogeneous PDE diagnostics for targeted finite-amplitude checks.

S6. Ecological-review sensitivity diagnostics

Figure 18 reports a finite-time sensitivity diagnostic for the defense-evolution rate ν. These runs use the same ODE equations and baseline parameterization except for ν, start from the pre-stress branch initial state, and test stresses $s \in \{0.08, 0.10, 0.1191, 0.14, 0.16\}$. The diagnostic is not a new rescue-window claim. It shows that finite-time rescue can depend on evolutionary rate even though the branch equilibrium itself is independent of ν. In particular, the slowest tested case loses predators at the representative rescue stress before adaptation completes, while the baseline $\nu = 0.05$ remains predator-persistent.

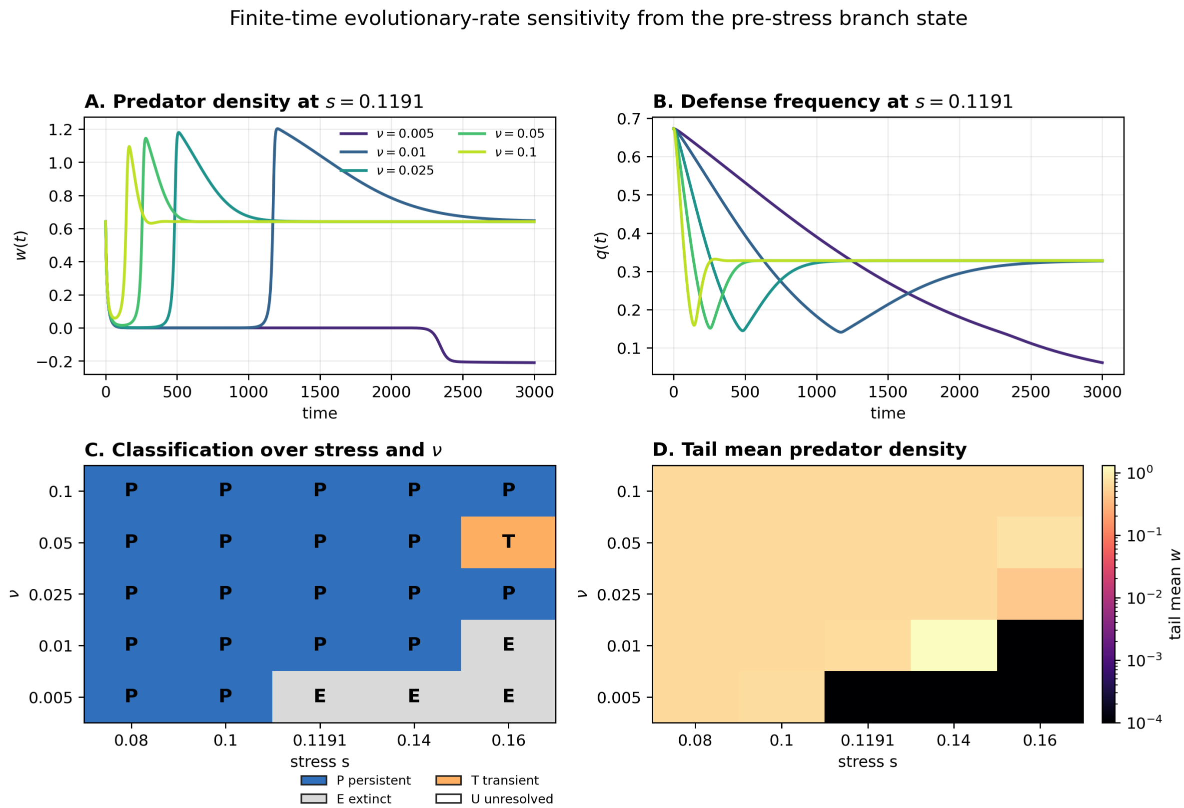

Figure 18: Finite-time evolutionary-rate sensitivity diagnostic. Panels A and B show $w(t)$ and $q(t)$ at $s = 0.1191$; Panels C and D summarize classifications and tail mean predator density across the small stress–ν grid.

Figure 19 reports a linear modal sensitivity diagnostic for the phenomenological q-diffusion coefficient. The calculation varies $D_q \in \{0, 0.001, 0.005, 0.01, 0.05\}$, keeps $D_n = D_w = 0.01$, and evaluates the same sampled continuous-λ range used in the main spatial-stability evidence. This is a linear diagnostic only; it does not add PDE simulations or establish finite-amplitude domain robustness.

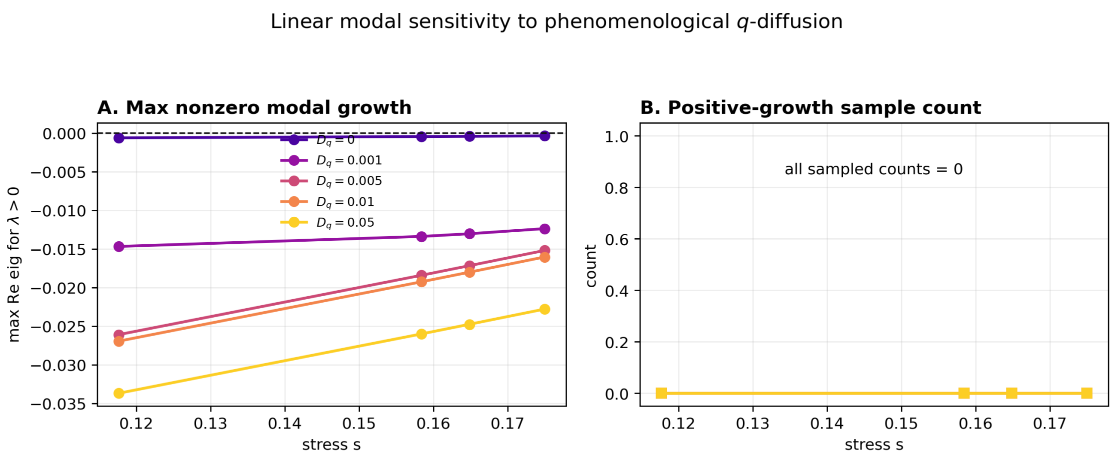

Figure 19: Linear modal sensitivity to q-diffusion. Panel A shows the maximum sampled growth rate for positive λ. Panel B reports the number of sampled positive-growth values.

S7. Additional figure list and CSV source mapping

Table 3 maps the main analysis scripts to the CSV outputs used by the manuscript package. Table 4 maps the main claims to source CSVs and representative figures. The status column records whether each item is a main result or supporting diagnostic evidence. All script names in these tables are relative to `experiments/`; all CSV names are relative to `results/`. For readability, the tables use numbered script short names and compact CSV names; the full paths are present in the repository.

Table 3: Script-to-CSV mapping for reproducibility.

Script	CSV outputs used here
Step 23: compensation conditions	`compensation_conditions_summary`; `compensation_condition_grid`
Step 24: Routh–Hurwitz	`routh_hurwitz_current`; `routh_hurwitz_summary`
Step 25: PDE spatial and heterogeneous tests	`pde_spatial_modes_current`; `pde_spatial_nonlinear_decision`
Step 26: long-horizon basin-resolution analysis	`pde_long_horizon_decision`; summary and time-series CSVs
Step 27: nonlinear trade-offs	`nonlinear_compensation_decision`; `nonlinear_shape_summary`; `nonlinear_branch_recovery`
Step 28: counterfactual and modal scan	`no_evolution_counterfactual_summary`; stress-sweep and time-series CSVs; `pde_lambda_scan_summary`; `rh_margins`
Step 29: manuscript audit	`compensation_branch_value_audit`; `compensation_branch_reference_doi_audit`; `nonlinear_tradeoff_failure_modes`
Step 30: ecological-review sensitivities	`ode_nu_sensitivity_summary`; `pde_dq_modal_sensitivity_summary`

Table 4: Claim-to-source mapping for the manuscript package.

Claim	Source CSV	Representative figure	Status
Frozen-q counterfactual verifies indirect rescue over a finite stress window	`no_evolution_counterfactual_summary`; stress-sweep CSV	Main no-evolution figure	Main claim
Linear compensation branch conditions are satisfied in the current parameterization	`compensation_conditions_summary`; `compensation_branch_value_audit`	Main branch figure; Fig. 4	Main claim
Routh–Hurwitz stability holds at target stresses	`routh_hurwitz_summary`	Main Routh–Hurwitz figure; Fig. 1	Main claim
Spatial modes are stable for the current PDE coefficients over the tested range	`pde_spatial_nonlinear_decision`; `pde_lambda_scan_summary`	Main spatial-stability figure; Fig. 5; Fig. 6	Main claim
Long-horizon non-homogeneous basin changes resolve to homogeneous controls	`pde_long_horizon_decision`	Main long-horizon decision panel; Fig. 11; Fig. 12; Fig. 13	Main claim
Finite-time rescue can depend on ν	`ode_nu_sensitivity_summary`	Fig. 18	Supporting diagnostic
Linear q-diffusion sensitivity shows no positive sampled modal growth in the tested set	`pde_dq_modal_sensitivity_summary`	Fig. 19	Supporting diagnostic
Linear branch is recovered by nonlinear branch formulation	`nonlinear_branch_recovery`	Fig. 14	Supporting
Nonlinear shape grid includes robust, partial, no-compensation, and unresolved cases	`nonlinear_shape_summary`; `nonlinear_tradeoff_failure_modes`; `compensation_branch_value_audit`	Main nonlinear heatmap; Table 2; Fig. 14	Main claim

The main manuscript contains the primary compensation branch and stress-interval figure. This boxed note avoids duplicating that main-text figure in the supplement while keeping the claim-to-source map readable.

Figure 20: Pointer to the main compensation branch figure.